

Lecture 3 — Markov Decision Processes

Reinforcement Learning 2026–27

(slides by Roberto Capobianco; companion text: book chapter 4)

Lecture notes

Structure follows the slide deck (slides 1–55); formal statements and proofs are merged in from the companion PDF at the point where the slides introduce the topic. Material that appears only in the PDF is collected in the Appendix.

Contents

1	Recap (<i>slides 2–11</i>)	1
1.1	From Multi-Armed to Contextual Bandits (<i>slides 3</i>)	1
1.2	Contextual Bandits: Interaction (<i>slides 4</i>)	1
1.3	Contextual Bandits: Regret (<i>slides 5</i>)	2
1.4	Explore & Commit Algorithm (<i>slides 6</i>)	2
1.5	ϵ -Greedy (<i>slides 7</i>)	2
1.6	Bayesian Bandits (<i>slides 8</i>)	3
1.7	Gaussian Bayesian Bandits: UCB (<i>slides 9</i>)	3
1.8	Gaussian Bayesian Bandits: Thompson Sampling (<i>slides 10</i>)	3
2	Sequential Decision Making (<i>slides 12–14</i>)	4
3	Markov Assumption (<i>slides 15–17</i>)	4
4	Markov Decision Process (MDP) (<i>slides 18–19</i>)	5
5	Reward (<i>slides 20</i>)	6
6	Deterministic MDP Example (<i>slides 21</i>)	6
7	Stochastic MDP Example (<i>slides 22</i>)	7
8	Policy (<i>slides 23</i>)	8
9	Trajectory Probability (<i>slides 24</i>)	8
10	State Visitation Probability (<i>slides 25–27</i>)	8
10.1	Definition (<i>slides 25</i>)	8
10.2	Examples (<i>slides 26–27</i>)	9
11	Another Example MDP (<i>slides 28</i>)	10
12	Infinite Horizon Discounted Setting (<i>slides 29</i>)	10
13	Value Function (<i>slides 30–31</i>)	10
14	Back to Discount Factor (<i>slides 32</i>)	11

15 Bellman Equation (<i>slides 33–36</i>)	11
16 Discounted State-Action Distribution (<i>slides 37–38</i>)	12
17 Optimal Policy (<i>slides 39–40</i>)	13
18 Bellman Optimality (<i>slides 41–42</i>)	13
19 Bellman Optimality Example (<i>slides 43–45</i>)	14
20 Bellman Optimality (Theorem 1) (<i>slides 46–49</i>)	14
21 Bellman Optimality (Theorem 2) (<i>slides 50–55</i>)	15
22 Summary	16
A Appendix: supplementary material (in the companion PDF, not on the slides)	17
A.1 Discounted sums as expectations under $d_{s_0}^\pi$	17
A.2 Bellman operators (notation used in the PDF’s proofs)	18

1 Recap (*slides 2–11*)

The lecture opens by recalling the bandit material of the previous lectures (slide 2 opens the recap, slide 11 closes it). The point of the recap is to set up the jump from *one-step* decision making to *sequential* decision making.

1.1 From Multi-Armed to Contextual Bandits (*slides 3*)

A multi-armed bandit (MAB) is stateless: the agent picks an action and receives a reward. A *contextual bandit* (CB) adds back some context (a state): before acting, the agent sees a context x_t , and the reward depends on both the context and the arm.

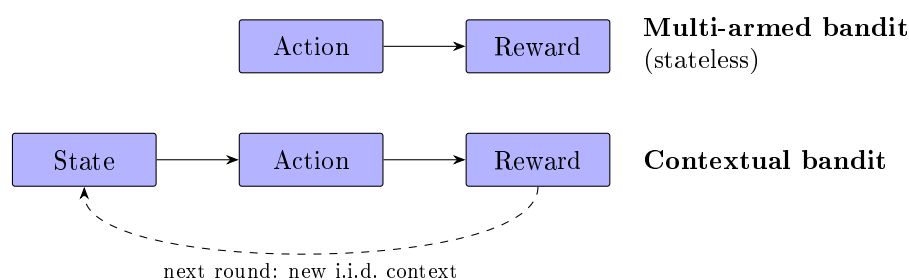


Figure 1: MAB versus contextual bandit (redrawn from slide 3).

Intuition

Think of a MAB as a single slot machine room with no information about who walks in; a CB is the same room where each customer has a profile x_t and the best machine depends on the profile. The crucial point for what follows: in a CB the context is drawn i.i.d., so *my action does not change the next context*. MDPs remove exactly this restriction.

1.2 Contextual Bandits: Interaction (*slides 4*)

For $t = 0, \dots, T - 1$:

1. a new i.i.d. context $x_t \in \mathcal{X}$ appears;
2. select an action $a_t \in \mathcal{A}$ based on historical information and the context;
3. observe the reward $r(x_t, a_t)$ (context- and arm-dependent).

For simplicity the rewards are assumed deterministic, “as the context is the challenge here”.

1.3 Contextual Bandits: Regret (*slides 5*)

The optimal policy is $\pi^* = \arg \max_{\pi \in \Pi} \mathbb{E}_{x \sim \mu} [r(x, \pi(x))]$. At every iteration $a_t = \pi^t(x_t)$ is chosen, and the regret compares the total expected reward of always using π^* with that of our learned sequence of policies:

$$\text{Regret}_T = T \mathbb{E}_{x \sim \mu} [r(x, \pi^*(x))] - \sum_{t=0}^{T-1} \mathbb{E}_{x \sim \mu} [r(x, \pi^t(x))].$$

Note that the policies π^t are different at every iteration t .

Intuition

Regret is “how much reward I left on the table by not knowing π^* from day one”. If Regret_T grows sublinearly in T , the *average* regret per round goes to 0: eventually the learner behaves like the optimal policy.

1.4 Explore & Commit Algorithm (*slides 6*)

Algorithm 1.1 (Explore & Commit for contextual bandits). 1. For $t = 0, \dots, N-1$ (*explore*): observe $x_t \sim \mu$; sample $a_t \sim \text{Unif}(\mathcal{A})$; observe $r_t = r(x_t, a_t)$; build, for x_t , an unbiased estimate $\hat{r}_t$ of the reward vector, with $\mathbb{E}_{a_t \sim \text{Unif}(\mathcal{A})} \hat{r}_t[a] = r(x_t, a) \forall a$:

$$\hat{r}_t[a] = \begin{cases} 0 & a \neq a_t \\ \frac{r_t}{1/|\mathcal{A}|} & a = a_t. \end{cases}$$

2. Compute the policy $\hat{\pi} = \arg \max_{\pi \in \Pi} \sum_{t=0}^{N-1} \hat{r}_t[\pi(x_t)]$.
3. For $t = N, \dots, T - 1$ (*commit*): observe $x_t \sim \mu$ and play $a_t = \hat{\pi}(x_t)$.

With a suitable N the regret is

$$\text{Regret}_T = O\left(T^{2/3} K^{1/3} \ln(|\Pi|)^{1/3}\right), \quad K = |\mathcal{A}|.$$

Intuition

The division by $1/|\mathcal{A}|$ is *importance weighting*: we only see the reward of the arm we actually played, which happens with probability $1/|\mathcal{A}|$, so we inflate it by $|\mathcal{A}|$ to be right on average. A numerical feel for the rate: doubling the horizon multiplies the regret bound by only $2^{2/3} \approx 1.59$, so the per-round regret decays like $T^{-1/3}$.

1.5 ϵ -Greedy (*slides 7*)

Instead of fixing a threshold for exploring and then committing, interleave exploration and exploitation. For $t = 0, \dots, T$: observe $x_t \sim \mu$; draw

$$a_t \sim p_t = (1 - \epsilon) \delta(\pi^t(x_t)) + \epsilon \text{Unif}(\mathcal{A});$$

observe $r_t = r(x_t, a_t)$; build an unbiased estimate $\hat{r}_t$ for x_t (so that $\mathbb{E}_{a \sim p_t} \hat{r}_t[a] = r(x_t, a)$); update the policy $\pi^{t+1} = \arg \max_{\pi \in \Pi} \sum_{\tau \leq t} \hat{r}_\tau[\pi(x_\tau)]$. Here $\epsilon = 0$ means pure exploitation and $\epsilon = 1$ uniform exploration. (The slide only asserts unbiasedness; the usual choice is the importance-weighted estimator $\hat{r}_t[a] = \mathbb{I}[a = a_t] r_t / p_t(a_t)$.)

Intuition

Explore & Commit explores first, then stops learning. ϵ -greedy never stops: with $\epsilon = 0.1$, about one round in ten is a random experiment, and the other nine use the current best guess. It is robust to the case where the early estimates were wrong.

1.6 Bayesian Bandits (*slides 8*)

So far no assumption was made about the reward distributions ν_i ; only bounds on rewards were derived. In Bayesian bandits we (i) exploit *prior* knowledge of rewards, (ii) update a *posterior* distribution from historical information, and (iii) use the posterior to guide exploration, either through upper confidence bounds (Bayesian UCB) or probability matching (Thompson sampling).

1.7 Gaussian Bayesian Bandits: UCB (*slides 9*)

Now that a distribution is modelled, a notion of confidence comes for free: for Gaussians it is the standard deviation. The UCB rule picks

$$a_t = \arg \max_{i \in K} \mu_t(i) + \frac{c \sigma_t(i)}{\sqrt{N_t(i)}},$$

where $N_t(i)$ is the number of pulls of arm i and c a constant.

1.8 Gaussian Bayesian Bandits: Thompson Sampling (*slides 10*)

For $t = 0, \dots, T$: for each arm $i = 1, \dots, K$ sample $\tilde{r}_t(i)$ independently from $\mathcal{N}(\mu_{t-1}(i), \sigma_{t-1}^2(i))$; pull $I_t = \arg \max_i \tilde{r}_t(i)$; observe r_t ; update the posterior $p(\mu_t(i), \sigma_t^2(i) \mid r_t)$. This can be done with other distributions as well. The slide stresses that this is an estimate of the *reward*: in more general MDPs it is replaced by the Q function, i.e. we estimate a distribution over Q .

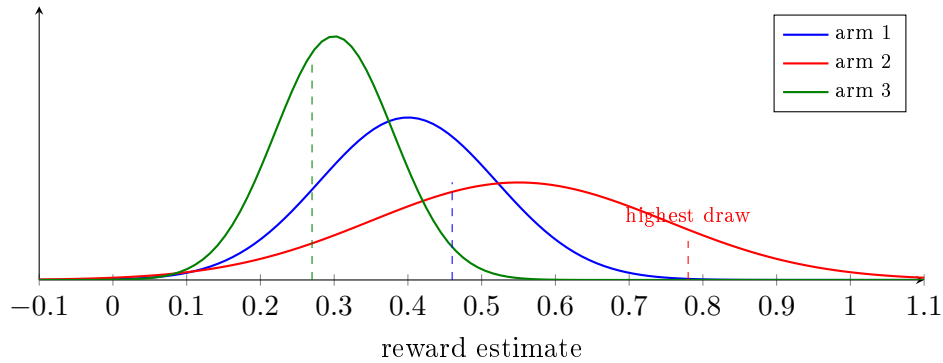


Figure 2: Thompson sampling with Gaussian posteriors (original illustration of the idea on slide 10; the numbers are illustrative). Dashed lines are the independent draws $\tilde{r}(i)$; the uncertain arm 2 gets the highest draw and is pulled.

Intuition

Thompson sampling explores in proportion to uncertainty without any explicit exploration parameter: an arm with a wide posterior occasionally produces a high draw and gets tried; once its posterior narrows and its mean is low, the draws stop beating the others. This is

the “probability matching” idea: pull each arm with the probability that it is the best.

2 Sequential Decision Making (*slides 12–14*)

The agent interacts with the environment at discrete time steps, by receiving an observation o_t and a reward r_t from the environment and by taking actions a_t (slide 12).

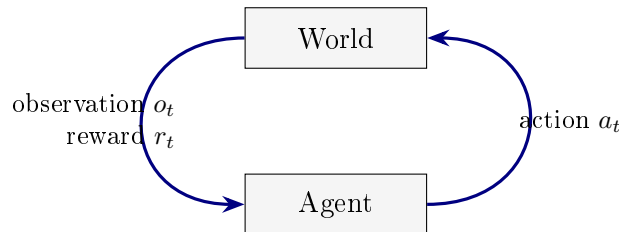


Figure 3: The agent–world loop (slides 12–14, redrawn).

Such a discrete interaction generates a trajectory, or *history*, at each time step t , which the agent uses to choose its action (slide 13):

$$h_t = (o_0, a_0, r_1, o_1, a_1, \dots, r_t, o_t, a_t) \quad (\text{indexing convention as on the slide}).$$

The *state* is a function of the history, $s_t = f(h_t)$, and it is typically hidden or unknown (slide 14).

Intuition

The history grows without bound; no practical agent can condition on all of it. The state is a *summary* of the history that is supposed to keep what matters for the future. Which summary is a good one is exactly the question the Markov assumption answers.

3 Markov Assumption (*slides 15–17*)

Definition 3.1 (Markov state). A state representation is Markov if for every t , every history h_t and every action a_t ,

$$p(s_{t+1} \mid s_t, a_t) = p(s_{t+1} \mid h_t, a_t).$$

The future is conditionally independent of the past given the present.

Intuition

If I tell you the current state and my action, nothing in the earlier history can sharpen your prediction of the next state. A chess position is Markov (how we reached it is irrelevant for what can happen next). The position of a moving ball alone is *not*: without its velocity, the past positions carry extra information.



Figure 4: Is this problem Markovian?

Slide 16 asks, over a picture of a robot in a corridor with two pillars, “Is this problem Markovian?” The answer depends on what we put in the state: if the state is only the robot’s position but the dynamics depend on, e.g., its velocity or an unobserved internal variable, past observations help predict the future and the state is *not* Markov; adding the missing variables restores the property.

Slide 17 gives the two extremes:

- a state can *always* be made Markov by taking it equal to the whole history, $s_t = h_t$;
- the best case, used in practice, is that the state is (a sufficient statistic of) the latest observation, $s_t = o_t$. The state is then said to be *fully observable*.

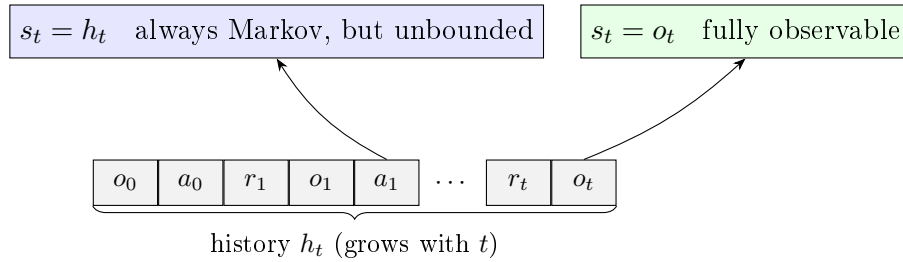


Figure 5: Two extreme choices of state as a function of the history (original illustration of slide 17).

Intuition

Choosing $s_t = h_t$ is a cheat: it is trivially Markov, but the state space explodes with time. Practical RL wants a small state that is still Markov. Fully observable means the observation already contains everything relevant; this is the setting of the rest of the lecture.

4 Markov Decision Process (MDP) (slides 18–19)

An MDP is sequential decision making under the Markov assumption: Markovian transition dynamics, full observability, and a (generally) stochastic transition kernel $p(s_{t+1} \mid s_t, a_t)$. Alternative notation: $s_{t+1} \sim p(\cdot \mid s_t, a_t)$ or $s' \sim p(\cdot \mid s, a)$ (slide 19).

Definition 4.1 (Markov decision process). A discounted infinite-horizon MDP is a tuple $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$: a set of states, a set of actions, transition dynamics $p(s' \mid s, a)$ satisfying Definition 3.1, a reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow [0, R_{\max}]$, and a discount factor $\gamma \in [0, 1)$. Unless stated otherwise $\mathcal{S}$ and $\mathcal{A}$ are finite.

(The slides introduce γ later, in the “Infinite Horizon Discounted Setting” part; the tuple is stated here in full for reference.)

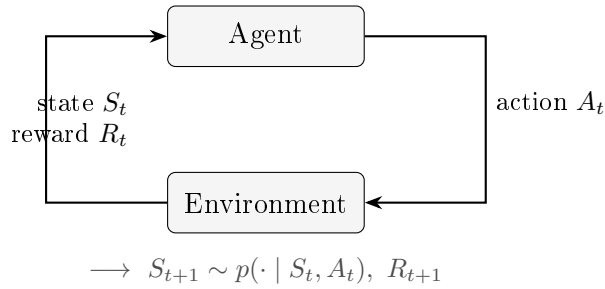


Figure 6: The MDP interaction loop (slides 18–19, redrawn).

Intuition

An MDP is the contextual bandit *with consequences*: the next “context” is the next state s' , and it is drawn from a distribution that depends on the action just taken. Every action therefore has two effects: the reward now, and the situation in which the next decision is made.

5 Reward (*slides 20*)

A reward r_t is a scalar signal representing feedback and indicating how well the agent is doing at step t . It is a function of state and action, written $r(s, a)$ (sometimes $r(s', a, s)$). A *cost* is the negative of a reward.

Reward hypothesis (slide 20, flagged as an open question): can all goals be achieved through the maximisation of a numerical reward?

Intuition

The reward is the only way we tell the agent what we want; everything else (what is a good state, a good action) must be *derived* from it. That is why designing rewards is delicate: the agent maximises what we write, not what we meant.

6 Deterministic MDP Example (*slides 21*)

The slide shows a small deterministic MDP of a developer’s day: states *Wake Up* ($r = 0$), *Netflix* ($r = -2$, with a self-loop), *Code & Debug* ($r = -3$), *Nap* ($r = 1$), *Deploy* ($r = 10$) and *Sleep* ($r = 3$, terminal).

Example 6.1 (Worked numbers, not on the slide). Take $\gamma = 0.9$, collect the reward of a state when leaving it, and treat Sleep as terminal (its reward is collected once). Then:

- $V(\text{Sleep}) = 3$, $V(\text{Deploy}) = 10 + 0.9 \cdot 3 = 12.7$, $V(\text{Netflix}) = -2/(1 - 0.9) = -20$ (it can only loop).
- From Code & Debug: $Q(\text{Code}, \text{Deploy}) = -3 + 0.9 \cdot 12.7 = 8.43$, $Q(\text{Code}, \text{Sleep}) = -3 + 0.9 \cdot 3 = -0.3$. The best choice is Deploy: a “bad” step (reward -3) pays off.
- Hence the path Wake→Code→Deploy→Sleep has discounted return $0 + 0.9 \cdot 8.43 = 7.587$, while Wake→Netflix forever has return $0 + 0.9 \cdot (-20) = -18$.
- Going Code→Nap→Wake gives $V(\text{Nap}) = 1 + 0.9 \cdot 7.587 \approx 7.83$ and $Q(\text{Code}, \text{Nap}) \approx 4.05 < 8.43$, so Nap loses.

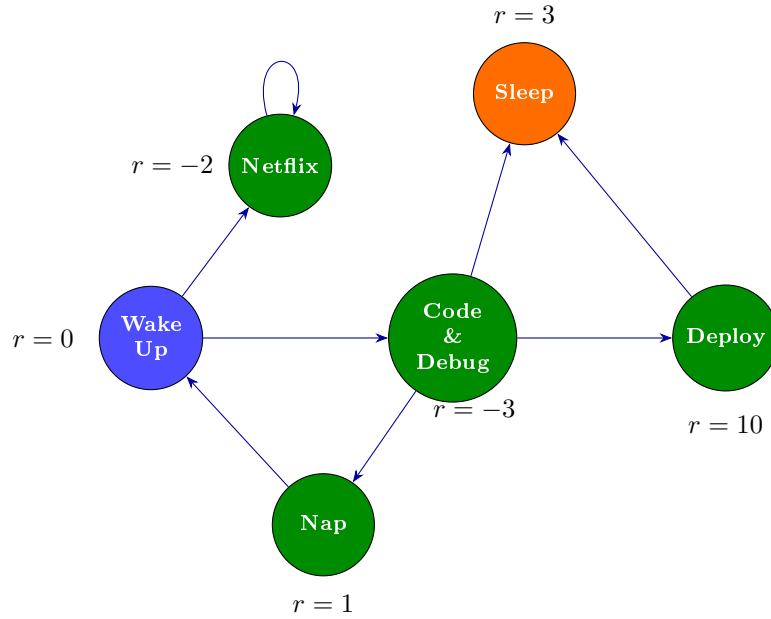


Figure 7: The deterministic MDP of slide 21, redrawn. Rewards are attached to states.

Intuition

The immediate reward at Code & Debug is negative (-3), yet it is the right place to go through: all the value sits behind it (Deploy = 10). A myopic agent at Wake Up compares -2 (Netflix) with -3 (Code & Debug) and picks Netflix, where it is stuck forever with value -20 ; maximising the *return* sends it through Code & Debug instead. This is the entire reason we define value functions.

7 Stochastic MDP Example (*slides 22*)

The slide (taken from Sutton & Barto) shows the *recycling robot*: states **high** and **low** battery; actions **search**, **wait**, **recharge**. The transition table of the slide is:

s	a	s'	$p(s' s, a)$	$r(s, a, s')$
high	search	high	α	r_{search}
high	search	low	$1 - \alpha$	r_{search}
low	search	high	$1 - \beta$	-3
low	search	low	β	r_{search}
high	wait	high	1	r_{wait}
high	wait	low	0	$-$
low	wait	high	0	$-$
low	wait	low	1	r_{wait}
low	recharge	high	1	0
low	recharge	low	0	$-$

Example 7.1 (Worked numbers, not on the slide). Let $\alpha = 0.7$, $\beta = 0.4$, $r_{\text{search}} = 2$, $r_{\text{wait}} = 1$. At **low**, the expected *immediate* reward of **search** is $0.4 \cdot 2 + 0.6 \cdot (-3) = -1.0$, versus $+1$ for **wait**. Searching with a low battery is risky: with probability 0.6 the robot runs flat, is rescued (reward -3) and ends up at **high**.

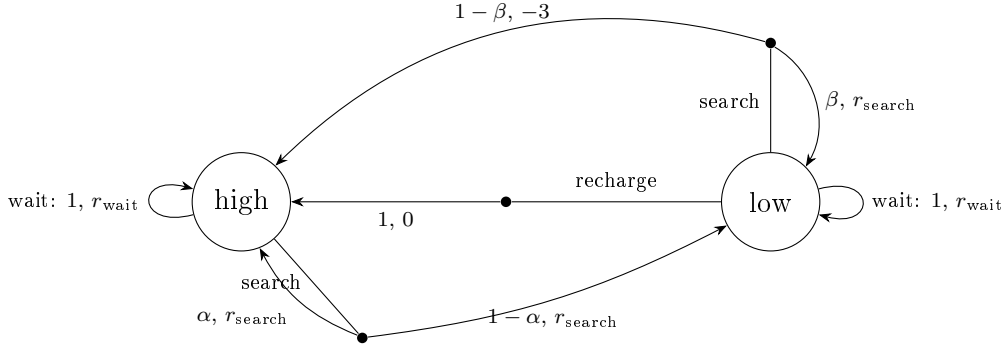


Figure 8: Transition graph of the recycling robot (slide 22, redrawn). Black dots are actions; edge labels are (probability, reward).

Intuition

Stochasticity means the same action from the same state can lead to different next states. Here a single decision (“search with a low battery”) is a gamble, and the right choice depends on the state: at **high** searching is cheap, at **low** it is not. A policy is precisely a rule for making this state-dependent choice.

8 Policy (*slides 23*)

A policy π is a mapping from (all) states to actions that determines how the agent selects actions. It can be deterministic, $a = \pi(s)$, or stochastic, $\pi(a | s)$, $p(a | s)$ or $a \sim \pi(\cdot | s)$.

Intuition

A policy is a complete *strategy*, not a plan: it says what to do in *every* state, including the ones I might never reach. In the recycling robot there are $2 \cdot 3 = 6$ deterministic policies (two choices at **high**, three at **low**); in general there are $|\mathcal{A}|^{|\mathcal{S}|}$, which is why we will need structure (Bellman equations) rather than brute-force enumeration.

9 Trajectory Probability (*slides 24*)

What is the probability of seeing the trajectory $(s_0, a_0, s_1, a_1, \dots, s_t, a_t)$ at time t when following π from s_0 ? Using the Markov property and the fact that π depends only on the current state:

$$\mathbb{P}^\pi(s_0, a_0, \dots, s_t, a_t) = \pi(a_0 | s_0) p(s_1 | s_0, a_0) \pi(a_1 | s_1) \cdots p(s_t | s_{t-1}, a_{t-1}) \pi(a_t | s_t).$$

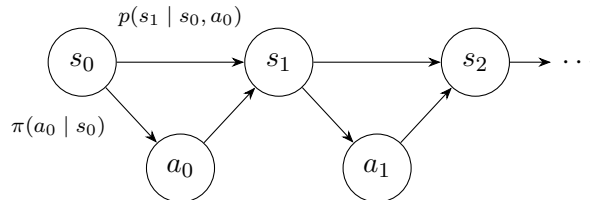


Figure 9: Graphical model of a trajectory (slide 24, redrawn).

Intuition

Each factor is one coin flip: the agent's flip (π) followed by the world's flip (p). Numbers: if $\pi(a_0|s_0) = 0.5$, $p(s_1|s_0, a_0) = 0.9$ and $\pi(a_1|s_1) = 1$, then the probability of the length-2 trajectory (s_0, a_0, s_1, a_1) is $0.5 \cdot 0.9 \cdot 1 = 0.45$.

10 State Visitation Probability (*slides 25–27*)

10.1 Definition (*slides 25*)

What is the probability of visiting (s, a) at time t when following π from s_0 ? Marginalise the trajectory probability over everything except the last pair:

$$\mathbb{P}_t^\pi(s, a; s_0) = \sum_{a_0, s_1, a_1, \dots, s_{t-1}, a_{t-1}} \mathbb{P}^\pi(s_0, a_0, \dots, s_t = s, a_t = a).$$

Intuition

It answers “if I run the policy many times, in what fraction of the runs am I in (s, a) at time t ?” Plotting it for every state gives a heat map that moves and spreads with t .

10.2 Examples (*slides 26–27*)

Slide 26 shows, for a policy that goes to the goal in a gridworld with 10% slip, the visitation probability estimated over 2000 simulations, together with the probability mass in the (absorbing) goal as a function of t . Slide 27 shows the same for a *random* policy, on a log scale: the mass at the goal stays essentially 0.

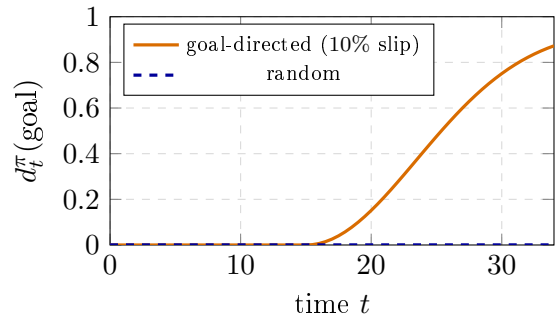
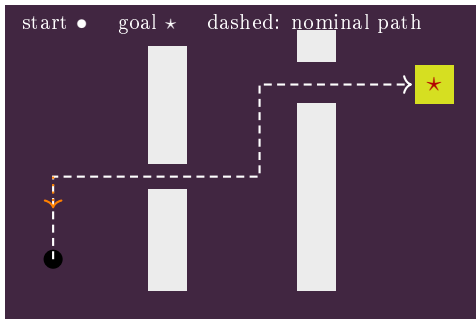


Figure 10: Schematic gridworld (left) and shape of the probability mass at the absorbing goal (right) for a goal-directed policy versus a random one. The curve is an original schematic read off slides 26–27, *not* the data of the slide.

For a policy that goes to the goal, 10% slip,
computed over 2000 simulations

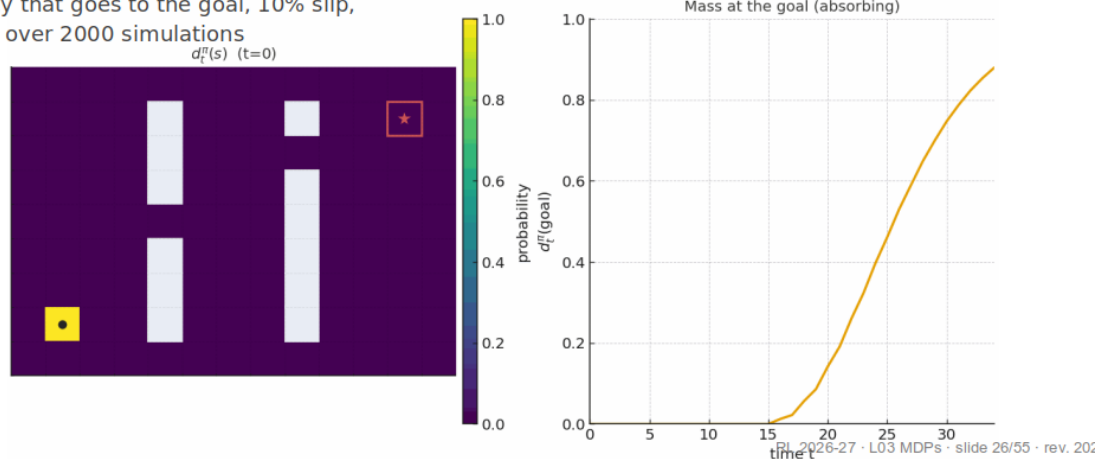


Figure 11: Heat map of $d_t^\pi(s)$ for the goal-directed policy (10% slip, 2000 simulations) with the “mass at the goal” curve.

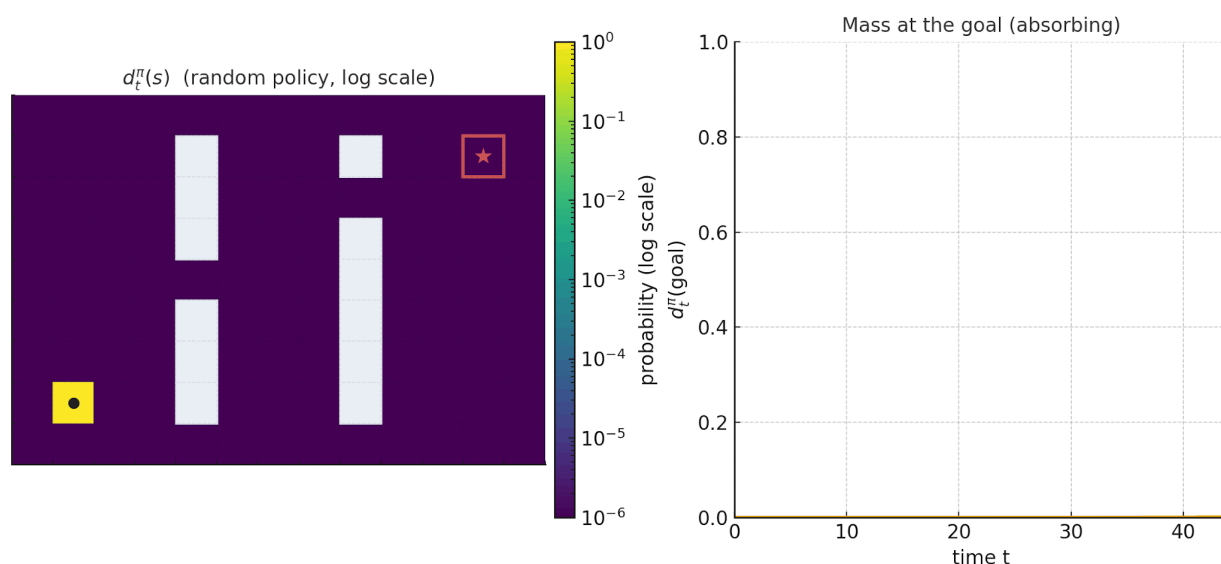


Figure 12: Same for the random policy, probability on log scale.

Intuition

Under the goal-directed policy nothing reaches the goal for the first ~ 15 steps (that is the minimum distance), then the mass climbs towards 1; the missing mass is runs that slipped and are still on the way. The random policy diffuses over the grid so thinly that its goal mass is almost 0 at every time: the visitation distribution is a fingerprint of the policy’s behaviour.

11 Another Example MDP (*slides 28*)

A robotic arm that must move a ball (slide 28):

- **state:** robot configuration (joint states) and ball position;
- **action:** torque on arm and finger joints;
- **transition:** stochastic, physics plus noise;
- **policy:** mapping from robot state and ball position to torque;
- **cost:** magnitude of the torque and distance to the goal.

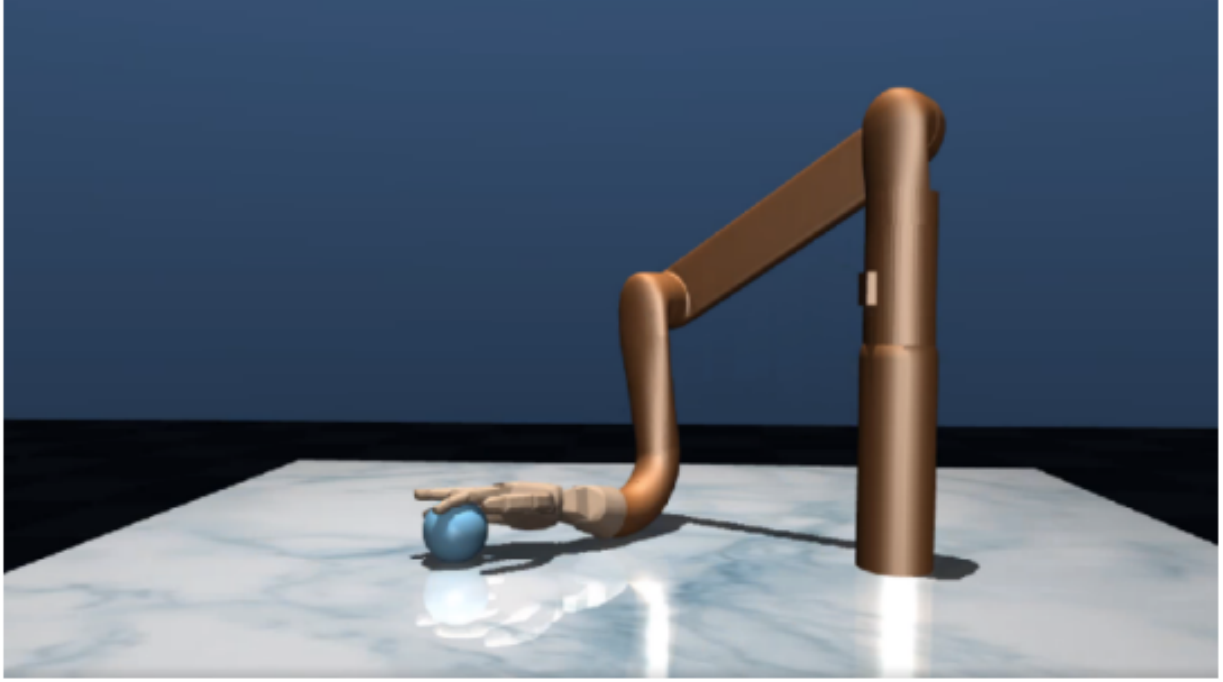


Figure 13: Photo of a robotic arm manipulating a ball

Intuition

Same five ingredients, a very different world: continuous states and actions, physics instead of a table of probabilities, and a cost made of two competing terms (effort and distance to goal). The abstract tuple $(\mathcal{S}, \mathcal{A}, p, r)$ covers both the toy table above and a robot.

12 Infinite Horizon Discounted Setting (*slides 29*)

So far the MDP is $(\mathcal{S}, \mathcal{A}, p, r)$. We now add a discount factor γ to reason about the long-term effects of a policy, with γ in $[0, 1]$ on the slide: $\gamma = 0$ means “I only care about immediate rewards”, while $\gamma = 1$ means immediate and future rewards are equally important. The slide leaves “How so?” as a question; it is answered in “Back to Discount Factor” below, where we see that $\gamma = 1$ is a bad idea for infinite tasks and the formal definition (Definition 4.1) takes $\gamma < 1$.

Intuition

γ is a patience dial. At $\gamma = 0.5$ a reward two steps away is worth 0.25 of the same reward now; at $\gamma = 0.99$ it is worth 0.98.

13 Value Function (*slides 30–31*)

We estimate the *goodness* of states and actions by their value, which is also a measure to compare policies. The state-value function of a policy π is the expected discounted return from s (slide 30):

$$\begin{aligned} V^\pi(s_t) &= \mathbb{E}[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots \mid s_t] \\ &= \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r_{t+h} \mid s_t = s, a_h = \pi(s_h), s_{h+1} \sim p(\cdot \mid s_h, a_h)\right]. \end{aligned}$$

Slide 31 adds the action-value function, where the first action is fixed and then π is followed:

$$Q^\pi(s_t, a_t) = \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h r_{t+h} \mid (s_t, a_t) = (s, a), \right. \\ \left. a_h = \pi(s_h), s_{h+1} \sim p(\cdot \mid s_h, a_h) \right].$$

Intuition

$V^\pi(s)$: “how good is it to *be* in s if I then follow π ?” $Q^\pi(s, a)$: “how good is it to take a *now* in s and then follow π ?” The difference between the two is one free first decision. Comparing $Q^\pi(s, a)$ across a is how an agent decides which action is better than the policy’s current choice.

14 Back to Discount Factor (*slides 32*)

Setting $\gamma = 1$ for infinite tasks is a bad idea. The quantity $\sum_{h=0}^{\infty} \gamma^h$ is a geometric series, equal to $1/(1 - \gamma)$ for $\gamma \in [0, 1)$, so the value of γ approximately determines how many steps ahead we are looking: e.g. $\gamma = 0.99$ corresponds to ~ 100 time steps.

Lemma 14.1 (Values are finite, and how finite). *For every policy and every state, $0 \leq V^\pi(s) \leq \frac{R_{\max}}{1 - \gamma}$.*

Proof. Rewards lie in $[0, R_{\max}]$, so $0 \leq \sum_h \gamma^h r_h \leq R_{\max} \sum_{h \geq 0} \gamma^h = R_{\max}/(1 - \gamma)$, the geometric series converging because $\gamma < 1$. Expectations preserve the bounds. $\square$

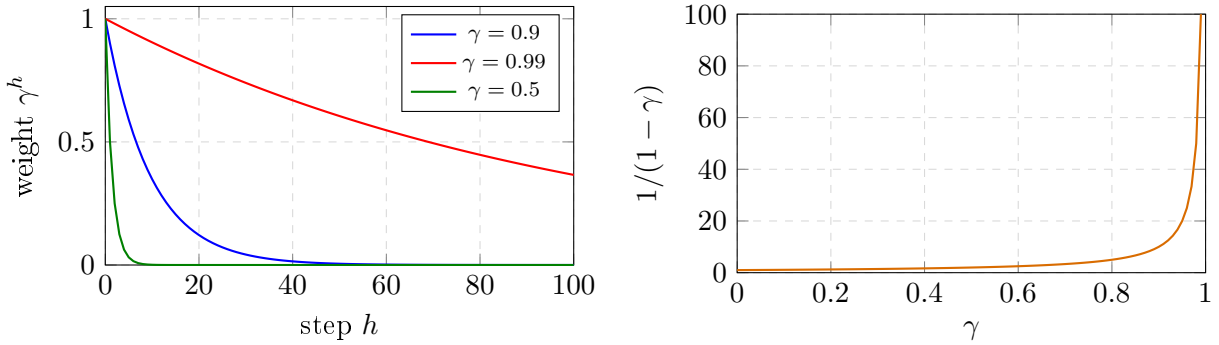


Figure 14: Left: discount weights γ^h . Right: the effective horizon $1/(1 - \gamma)$ diverges as $\gamma \rightarrow 1$ (original figures for slide 32).

Intuition

With $\gamma = 0.99$ the weight γ^h falls to $\gamma^{100} \approx 0.37 \approx 1/e$ after about 100 steps; with $\gamma = 0.9$ the same happens after about 10 steps. If rewards are at most $R_{\max} = 1$, the value can never exceed $1/(1 - 0.99) = 100$; at $\gamma = 1$ the bound is infinite and the sum of rewards of an endless task need not exist at all. The discount is what makes “the value of a state” a well-defined finite number.

15 Bellman Equation (*slides 33–36*)

The value of a state is expanded in terms of the current reward and the value of the next states according to the policy (slide 33): for a deterministic π ,

$$V^\pi(s_t) = \mathbb{E}_\pi[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots \mid s_t] = r_t + \gamma \mathbb{E}_{s' \sim p(\cdot \mid s_t, \pi(s_t))} [V^\pi(s')],$$

and likewise for Q (slides 34–36):

$$Q^\pi(s_t, a_t) = r_t + \gamma \mathbb{E}_{s' \sim p(\cdot \mid s_t, a_t)} [V^\pi(s')].$$

In the first equation r is a function of s and $\pi(s)$; in the second, of s and a . As a result $V^\pi(s) = Q^\pi(s, \pi(s))$ (slide 36).

Theorem 15.1 (Bellman expectation equations). *For every policy π and all s, a :*

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{a \sim \pi(\cdot \mid s)} [r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot \mid s, a)} [V^\pi(s')]], \\ Q^\pi(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot \mid s, a)} [V^\pi(s')], \\ V^\pi(s) &= \mathbb{E}_{a \sim \pi(\cdot \mid s)} [Q^\pi(s, a)]. \end{aligned}$$

For a deterministic policy the first reads $V^\pi(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim p(\cdot \mid s, \pi(s))} [V^\pi(s')]$ and the last $V^\pi(s) = Q^\pi(s, \pi(s))$.

Proof. Split the return after the first step using $G_0 = r(s_0, a_0) + \gamma G_1$: $V^\pi(s) = \mathbb{E}^\pi[G_0 \mid s_0 = s] = \mathbb{E}^\pi[r(s_0, a_0) \mid s_0 = s] + \gamma \mathbb{E}^\pi[G_1 \mid s_0 = s]$. Condition the second term on (a_0, s_1) and use the tower rule. By Definition 3.1 and the fact that π depends only on the current state, the conditional distribution of the trajectory from s_1 onwards given (s_0, a_0, s_1) is the same as that of a trajectory started at s_1 ; hence $\mathbb{E}^\pi[G_1 \mid s_0 = s, a_0 = a, s_1 = s'] = V^\pi(s')$. Averaging over $s' \sim p(\cdot \mid s, a)$ and then over $a \sim \pi(\cdot \mid s)$ gives the first identity; the argument for Q^π is the same with a_0 fixed. $\square$

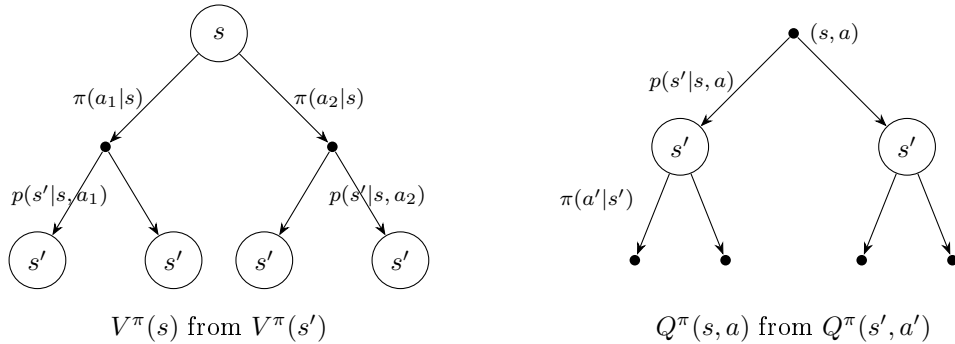


Figure 15: Backup diagrams for the Bellman expectation equations (original figure for slides 33–36).

Intuition

The equation says: *value now = reward now + discounted value of where I land*. It turns an infinite sum into a one-step consistency condition. Example (deterministic): if $r(s, \pi(s)) = 1$, $\gamma = 0.9$ and the next state always has value $V^\pi(s') = 10$, then $V^\pi(s) = 1 + 0.9 \cdot 10 = 10$. The two lines of the theorem differ only in what is averaged: V averages over my action choice (policy), Q has the first action already fixed and averages only over the environment's randomness.

16 Discounted State-Action Distribution (*slides 37–38*)

Definition 16.1 (Discounted state-action distribution).

$$d_{s_0}^\pi(s, a) = (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h \mathbb{P}_h^\pi(s, a; s_0).$$

For an initial distribution μ write $d_\mu^\pi(s, a) = \mathbb{E}_{s_0 \sim \mu}[d_{s_0}^\pi(s, a)]$.

Lemma 16.2 (It is a distribution, and that is the point of the $(1 - \gamma)$). $d_{s_0}^\pi$ is a probability distribution over $\mathcal{S} \times \mathcal{A}$.

Proof. Non-negativity is clear. Summing over (s, a) and exchanging the (absolutely convergent) sums, $\sum_{s,a} d_{s_0}^\pi(s, a) = (1 - \gamma) \sum_h \gamma^h \sum_{s,a} \mathbb{P}_h^\pi(s, a; s_0) = (1 - \gamma) \sum_h \gamma^h = 1$, since $\mathbb{P}_h^\pi(\cdot; s_0)$ is a distribution for each h . $\square$

This matches the slide’s remark: “this gives us a probability distribution (remember $\sum_h \gamma^h = 1/(1 - \gamma)$)”. The companion identity expressing discounted sums as expectations under $d_{s_0}^\pi$ is not on the slides and is given in Appendix A.1.

Intuition

Take the visitation probabilities of the previous section, weight time step h by γ^h and sum them: this is “how much total (discounted) time I spend in (s, a) ”. Without the factor $(1 - \gamma)$ the total mass would be $1/(1 - \gamma)$ (e.g. 100 for $\gamma = 0.99$); multiplying by $(1 - \gamma)$ rescales it to 1 so it can be used as a genuine probability distribution. Early time steps get most of the weight, which is the same patience dial as before.

17 Optimal Policy (*slides 39–40*)

Definition 17.1 (Optimal value and optimal policy). $V^*(s) = \sup_\pi V^\pi(s)$ and $Q^*(s, a) = \sup_\pi Q^\pi(s, a)$. A policy π^* is optimal if $V^{\pi^*}(s) = V^*(s)$ for every s .

Slides 39–40 state that for infinite-horizon MDPs there *always* exists a deterministic policy π^* such that

$$V^{\pi^*}(s) \geq V^\pi(s) \quad \forall s, \pi,$$

i.e. π^* dominates all other policies in each state; it returns optimal actions a^* , and $V^{\pi^*} = V^*$, $Q^{\pi^*} = Q^*$. The existence statement is proved constructively by Theorem 20.1 below.

Intuition

It is far from obvious that one policy can be best *in all states simultaneously*; in many optimisation problems the best choice for one objective hurts another. Here it is true, and it is also true that we never need randomness: a table of “best action in each state” suffices. This is what makes the search for optimal behaviour tractable.

18 Bellman Optimality (*slides 41–42*)

Theorem 18.1 (Bellman optimality equation).

$$V^*(s) = \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s, a)} V^*(s') \right],$$

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s, a)} \left[\max_{a'} Q^*(s', a') \right].$$

On slide 42 the bracket $r(s, a) + \gamma \mathbb{E}_{s'} V^*(s')$ is annotated as $Q^*(s, a)$, so that $V^*(s) = \max_a Q^*(s, a)$.

Intuition

The Bellman *expectation* equation averages over the policy's action; the Bellman *optimality* equation replaces that average by a max: the optimal agent simply takes the best action. All the difficulty of looking arbitrarily far into the future is hidden in $V^*(s')$: if I knew the optimal value of every neighbour, choosing the best action would be a one-step problem.

19 Bellman Optimality Example (*slides 43–45*)

Assume we know V^* at s' and s'' , the two successors of s reached by actions a' and a'' (slide 43). Try a' , get $r(s, a')$ and compute $Q^*(s, a') = r(s, a') + \gamma V^*(s')$; try a'' and compute $Q^*(s, a'') = r(s, a'') + \gamma V^*(s'')$ (slide 44). Then (slide 45)

$$V^*(s) = \max_{a', a''} \{Q^*(s, a'), Q^*(s, a'')\}.$$

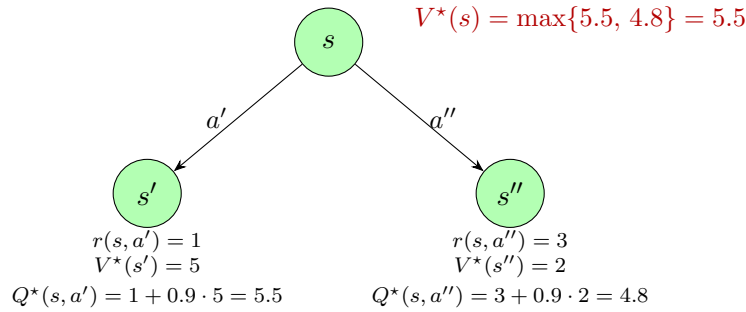


Figure 16: The one-step lookahead of slides 43–45 with illustrative numbers ($\gamma = 0.9$; the numbers are mine, not the slide's).

Intuition

With the numbers above the action with the *larger immediate reward* (a'' , reward 3) is *not* the best: a' leads to a state worth 5, and $1 + 4.5 > 3 + 1.8$. This is the one-step trick that everything else in the course rests on: evaluate each action as “reward + discounted value of where it leads” and take the best.

20 Bellman Optimality (Theorem 1) (*slides 46–49*)

Theorem 20.1 (Theorem 1 — greedy with respect to Q^* is optimal). *Let $\hat{\pi}(s) \in \arg \max_a Q^*(s, a)$. Then $V^{\hat{\pi}} = V^*$, and therefore $\pi^* = \hat{\pi}$ is an optimal policy.*

Proof as shown on the slides (slides 46–48)

Starting from $V^*(s) = \max_a [r(s, a) + \gamma \mathbb{E}_{s'} V^*(s')]$ and unrolling the equation along $\hat{\pi}$:

$$\begin{aligned} V^*(s) &= r(s, \pi^*(s)) + \gamma \mathbb{E}_{s'} V^*(s') \\ &\leq \max_a \left[r(s, a) + \gamma \mathbb{E}_{s'} V^*(s') \right] = r(s, \hat{\pi}(s)) + \gamma \mathbb{E}_{s'} V^*(s') \\ &\leq r(s, \hat{\pi}(s)) + \gamma \mathbb{E}_{s' \sim p(\cdot|s, \hat{\pi}(s))} \left[r(s', \hat{\pi}(s')) + \gamma \mathbb{E}_{s''} V^*(s'') \right] \\ &\leq \dots \leq \mathbb{E} \left[r(s, \hat{\pi}(s)) + \gamma r(s', \hat{\pi}(s')) + \dots \right] = V^{\hat{\pi}}(s), \end{aligned}$$

so $V^{\hat{\pi}} \geq V^*$ (the leftover term $\gamma^k \mathbb{E} V^*(s_k) \rightarrow 0$ since V^* is bounded, Lemma 14.1); and $V^* \geq V^{\hat{\pi}}$ holds by definition of V^* as a supremum. Hence $V^{\hat{\pi}} = V^*$ and, by slide 49, $\pi^* = \arg \max_a Q^*(s, a)$ is an optimal policy.

Proof in the companion text (fixed-point argument)

Proof. By Theorem 18.1, $V^* = \mathcal{B}V^*$ and $Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s'} [V^*(s')]$ (the operators $\mathcal{B}, \mathcal{B}^{\hat{\pi}}$ are recalled in Appendix A.2). Since $\hat{\pi}(s)$ maximises $Q^*(s, \cdot)$,

$$(\mathcal{B}V^*)(s) = \max_a Q^*(s, a) = Q^*(s, \hat{\pi}(s)) = r(s, \hat{\pi}(s)) + \gamma \mathbb{E}_{s' \sim p(\cdot|s, \hat{\pi}(s))} [V^*(s')] = (\mathcal{B}^{\hat{\pi}}V^*)(s),$$

so V^* is a fixed point of $\mathcal{B}^{\hat{\pi}}$. By Theorem 15.1 so is $V^{\hat{\pi}}$, and both are bounded by Lemma 14.1.

A bounded fixed point of $\mathcal{B}^{\hat{\pi}}$ is unique. If u and v are two, then for every s

$$|u(s) - v(s)| = \gamma \left| \mathbb{E}_{s'} [u(s') - v(s')] \right| \leq \gamma \|u - v\|_{\infty},$$

because the rewards cancel, so $\|u - v\|_{\infty} \leq \gamma \|u - v\|_{\infty}$. The norm is finite and $\gamma < 1$, hence $u = v$. Therefore $V^{\hat{\pi}} = V^*$, and $\hat{\pi}$ attains the supremum of Definition 17.1 at every state. $\square$

Intuition

Theorem 1 is the license to *act*: once someone hands me Q^* , I do not need to plan any more; the single-step rule “pick the action with the largest $Q^*(s, \cdot)$ ” is already globally optimal. The slide proof says it as a sandwich: following $\hat{\pi}$ can’t be better than optimal ($V^{\hat{\pi}} \leq V^*$), and by unrolling the optimality equation it can’t be worse either ($V^{\hat{\pi}} \geq V^*$). The book proof says the same thing differently: V^* and $V^{\hat{\pi}}$ solve the same one-policy equation, and that equation has only one bounded solution because each application shrinks distances by $\gamma < 1$.

21 Bellman Optimality (Theorem 2) (slides 50–55)

Theorem 21.1 (Theorem 2 — the optimality equation has one solution). *If $V : \mathcal{S} \rightarrow \mathbb{R}$ satisfies $V(s) = \max_a [r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot|s, a)} V(s')]$ for every s , then $V = V^*$.*

The slides (51–53) ask us to check that $\|V - V^*\|_{\infty} = 0$ and then unroll the difference.

Proof. Both V and V^* satisfy the same equation, so for every s ,

$$\begin{aligned} |V(s) - V^*(s)| &= \left| \max_a [r(s, a) + \gamma \mathbb{E}_{s'} V(s')] - \max_a [r(s, a) + \gamma \mathbb{E}_{s'} V^*(s')] \right| \\ &\leq \max_a \left| [r(s, a) + \gamma \mathbb{E}_{s'} V(s')] - [r(s, a) + \gamma \mathbb{E}_{s'} V^*(s')] \right| \\ &= \max_a \gamma \left| \mathbb{E}_{s' \sim p(\cdot|s, a)} [V(s') - V^*(s')] \right| \\ &\leq \gamma \max_a \mathbb{E}_{s' \sim p(\cdot|s, a)} |V(s') - V^*(s')|. \end{aligned}$$

The first inequality is $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$; the last is Jensen. Applying the same bound to $|V(s') - V^*(s')|$ and iterating k times,

$$|V(s) - V^*(s)| \leq \gamma^k \max_{a_0, \dots, a_{k-1}} \mathbb{E}_{s_k} |V(s_k) - V^*(s_k)| \leq \gamma^k \|V - V^*\|_\infty,$$

and since $\gamma < 1$ and $\|V - V^*\|_\infty$ is finite, the right-hand side tends to 0. Hence $|V(s) - V^*(s)| = 0$ for every s . $\square$

Slide 54 illustrates the shrinking bound $\gamma^h \|V - V^*\|_\infty$ as the number of future steps h grows (its example uses $\gamma = 0.92$ and a baseline $\|V - V^*\|_\infty = 1$).

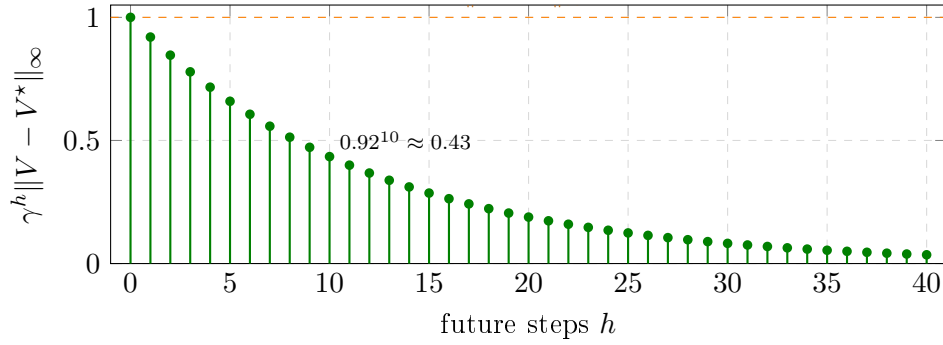


Figure 17: The bound $\gamma^h \|V - V^*\|_\infty$ for $\gamma = 0.92$, redrawn from the idea of slide 54 (values computed here).

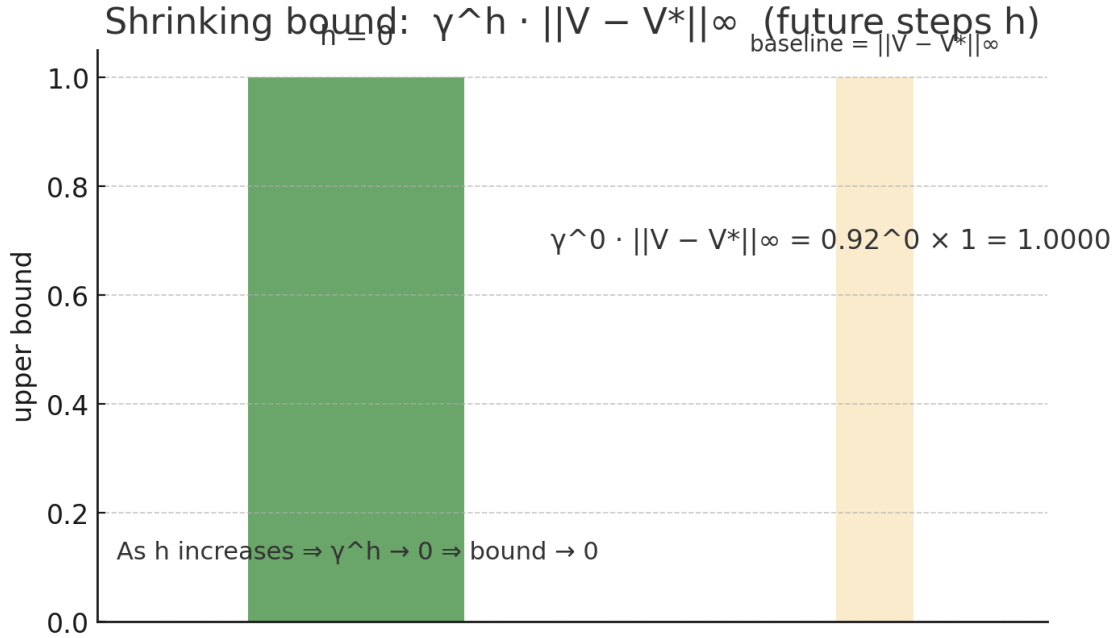


Figure 18: Shrinking bound

Intuition

Imagine a “fake” value function V that satisfies the optimality equation and is not V^* . Each time we expand the equation one step, the gap can be at most γ times what it was; after k

steps it is at most γ^k times the original gap, which melts away (with $\gamma = 0.92$, $0.92^{56} < 0.01$). The only way for the gap to survive all expansions is to be 0. Slide 55 states the consequence: *we can focus on one step at a time*, leaving the rest of the problem to $V(s')$, and any V satisfying that one-step formula is in fact V^* .

22 Summary

Object / result	Formula	What it tells us	Slides
Markov state	$p(s_{t+1} s_t, a_t) = p(s_{t+1} h_t, a_t)$	the present summarises the past	15–17
MDP	$(\mathcal{S}, \mathcal{A}, p, r, \gamma)$	sequential decisions with consequences	18–29
V^π, Q^π	$\mathbb{E}[\sum_h \gamma^h r_{t+h} \mid \cdot]$	how good a state / action is under π	30–31
Finite values	$0 \leq V^\pi \leq R_{\max}/(1 - \gamma)$	why $\gamma < 1$	32
Bellman expectation	$V^\pi(s) = \mathbb{E}_a[r + \gamma \mathbb{E}_{s'} V^\pi(s')]$	one-step consistency of a given policy	33–36
$d_{s_0}^\pi$	$(1 - \gamma) \sum_h \gamma^h \mathbb{P}_h^\pi$	discounted occupancy, a true distribution	37–38
Optimal V^*, Q^*	$\sup_\pi V^\pi, \sup_\pi Q^\pi$	a deterministic policy dominates all others	39–40
Bellman optimality	$V^*(s) = \max_a[r + \gamma \mathbb{E}_{s'} V^*(s')]$	one-step lookahead with a max	41–45
Theorem 1	$\hat{\pi} \in \arg \max_a Q^* \Rightarrow V^{\hat{\pi}} = V^*$	greedy w.r.t. Q^* is optimal	46–49
Theorem 2	$V = \max_a[r + \gamma \mathbb{E}V] \Rightarrow V = V^*$	V^* is the unique solution	50–55

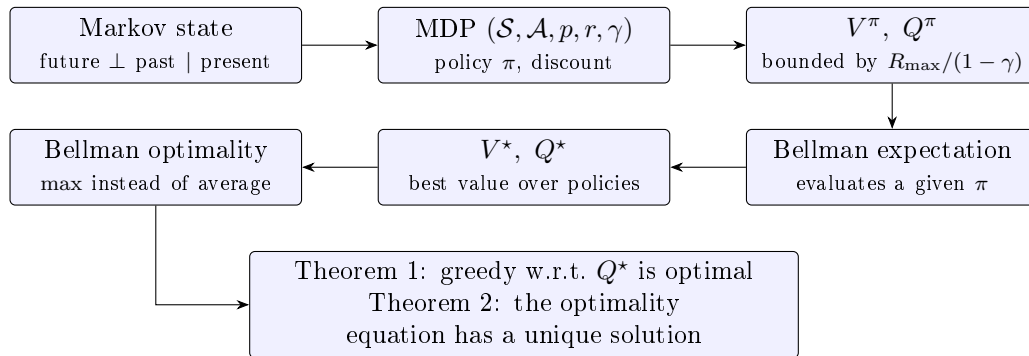


Figure 19: Roadmap of the lecture: from the Markov assumption to optimal policies.

Key take-aways

- An MDP is a contextual bandit whose next context depends on the action taken; the Markov property means the current state is all the agent needs to know.
- The discount factor $\gamma < 1$ keeps the value finite, with $0 \leq V^\pi \leq R_{\max}/(1 - \gamma)$, and $1/(1 - \gamma)$ is the effective horizon ($\gamma = 0.99 \Rightarrow \sim 100$ steps).
- Bellman equations turn an infinite-horizon sum into a one-step relation: value now = reward + discounted value of the next state (average for a policy, max for optimality).

- Theorem 1: acting greedily with respect to Q^* is optimal, so a deterministic stationary policy always suffices.
- Theorem 2: V^* is the *only* solution of the Bellman optimality equation, thanks to the γ -contraction; any V that satisfies the one-step formula is V^* .

A Appendix: supplementary material (in the companion PDF, not on the slides)

The following results appear in the companion PDF but have no corresponding slide; they are *not* part of the slide flow above.

A.1 Discounted sums as expectations under $d_{s_0}^\pi$

Lemma A.1 (Second part of Lemma 5 of the PDF). *For every bounded f ,*

$$\mathbb{E}^\pi \left[\sum_{h=0}^{\infty} \gamma^h f(s_h, a_h) \mid s_0 \right] = \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim d_{s_0}^\pi} [f(s, a)].$$

Proof. It is the same exchange of (absolutely convergent) sums used in the proof of Lemma 16.2, applied to f : $\mathbb{E}^\pi [\sum_h \gamma^h f(s_h, a_h) \mid s_0] = \sum_h \gamma^h \sum_{s,a} \mathbb{P}_h^\pi(s, a; s_0) f(s, a) = \frac{1}{1-\gamma} \sum_{s,a} d_{s_0}^\pi(s, a) f(s, a)$. $\square$

Intuition

Choosing $f = r$ gives $V^\pi(s_0) = \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim d_{s_0}^\pi} [r(s, a)]$: the value is the average reward under the discounted occupancy, scaled by the effective horizon $1/(1-\gamma)$. For $\gamma = 0.9$, if the average reward under d^π is 0.5, the value is $0.5/0.1 = 5$.

A.2 Bellman operators (notation used in the PDF's proofs)

The proof of Theorem 20.1 in the companion text uses two operators acting on bounded functions $V : \mathcal{S} \rightarrow \mathbb{R}$:

$$\begin{aligned} (\mathcal{B}V)(s) &= \max_a \left[r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, a)} V(s') \right], \\ (\mathcal{B}^\pi V)(s) &= r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, \pi(s))} V(s') \quad (\pi \text{ deterministic}). \end{aligned}$$

With this notation Theorem 18.1 reads $V^* = \mathcal{B}V^*$, Theorem 15.1 reads $V^\pi = \mathcal{B}^\pi V^\pi$, and Theorem 21.1 says that $\mathcal{B}$ has a unique fixed point. The slides never name these operators.